```
# Name : Aishwarya Laxmikant Lonikar
# CS AI - D
# Roll No. 2
# Subject : Machine Learning
# PRN : 12410541
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

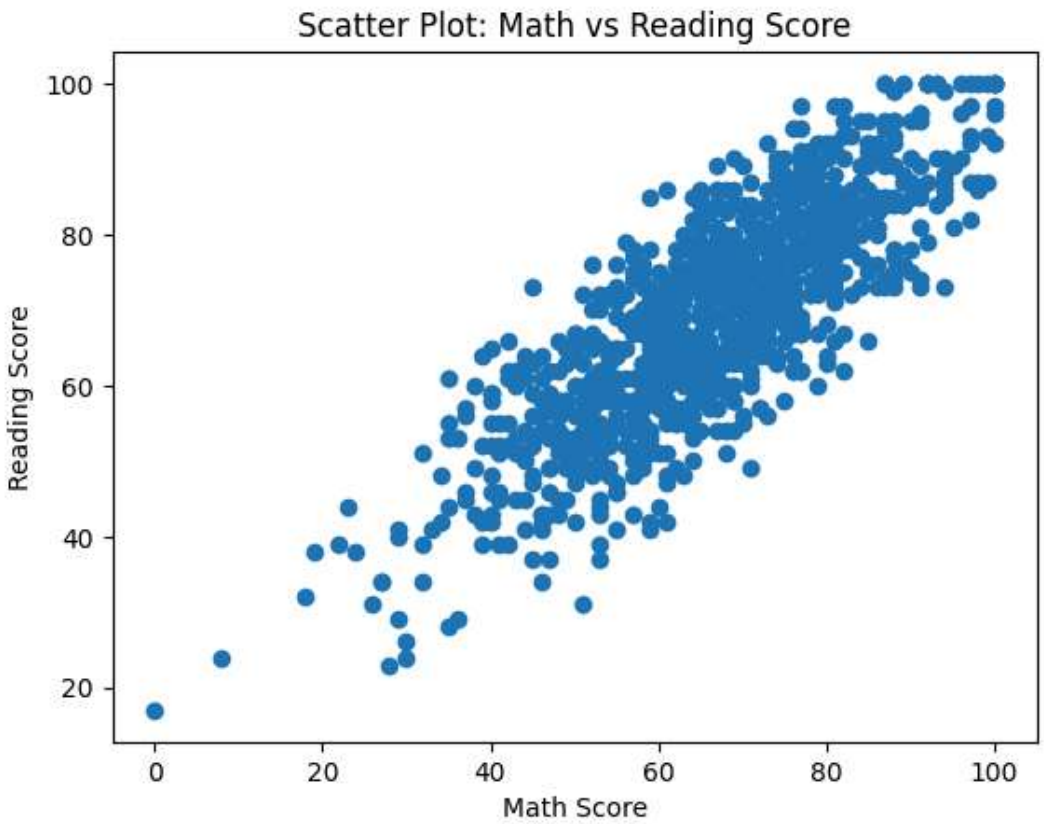
```
df_csv= pd.read_csv("/content/sample_data/study_performance.csv")
df_csv
```

| | gender | race_ethnicity | parental_level_of_education | lunch | test_preparation_course | math_score | reading_score | writing_ |
|---|---|---|---|---|---|---|---|---|
| **0** | female | group B | bachelor's degree | standard | none | 72 | 72 | |
| **1** | female | group C | some college | standard | completed | 69 | 90 | |
| **2** | female | group B | master's degree | standard | none | 90 | 95 | |
| **3** | male | group A | associate's degree | free/reduced | none | 47 | 57 | |
| **4** | male | group C | some college | standard | none | 76 | 78 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | |
| **995** | female | group E | master's degree | standard | completed | 88 | 99 | |
| **996** | male | group C | high school | free/reduced | none | 62 | 55 | |
| **997** | female | group C | high school | free/reduced | completed | 59 | 71 | |
| **998** | female | group D | some college | standard | completed | 68 | 78 | |
| **999** | female | group D | some college | free/reduced | none | 77 | 86 | |

1000 rows × 9 columns

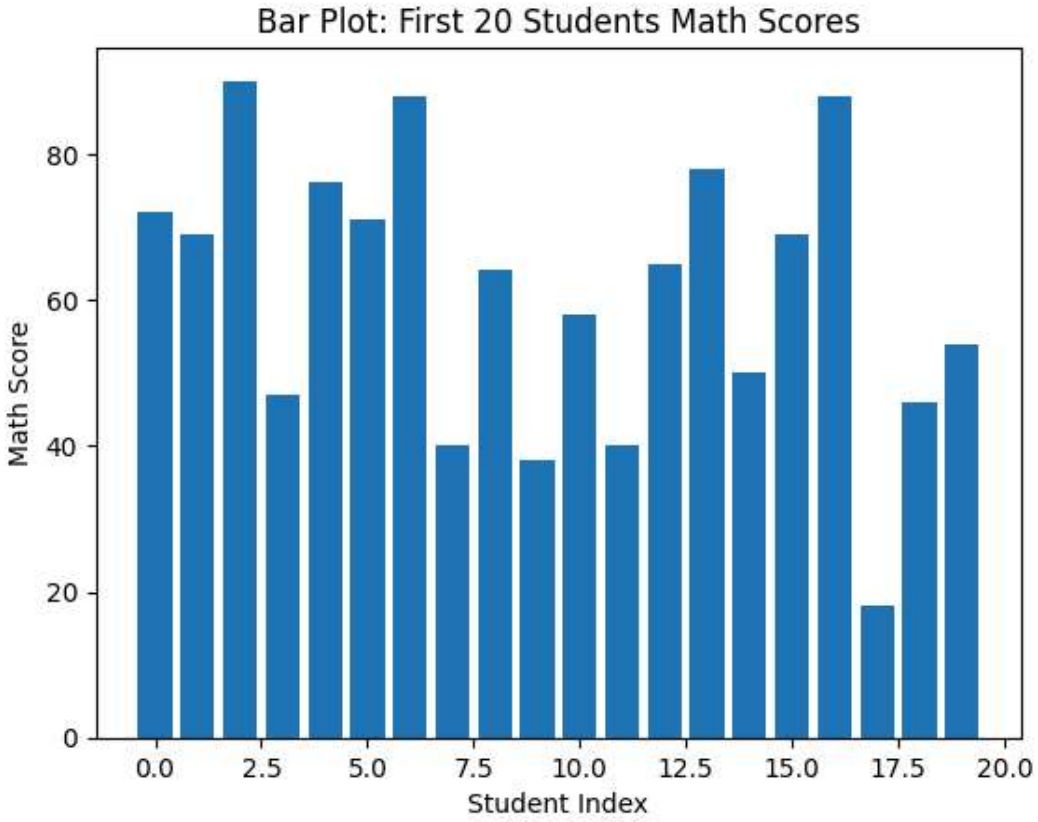
Next steps: New interactive sheet

```
plt.figure()
plt.scatter(df_csv['math_score'], df_csv['reading_score'])
plt.title("Scatter Plot: Math vs Reading Score")
plt.xlabel("Math Score")
plt.ylabel("Reading Score")
plt.show()
```

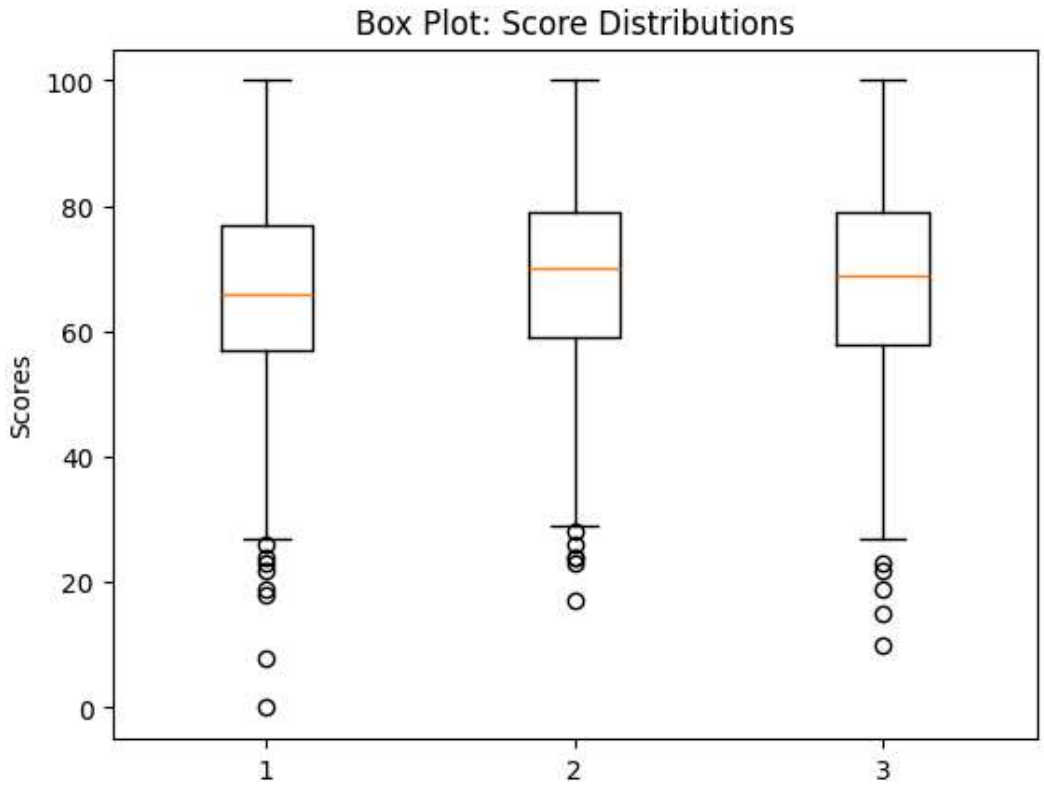



```
plt.figure()
plt.bar(range(len(df_csv['math_score'][:20])), df_csv['math_score'][:20])
plt.title("Bar Plot: First 20 Students Math Scores")
```

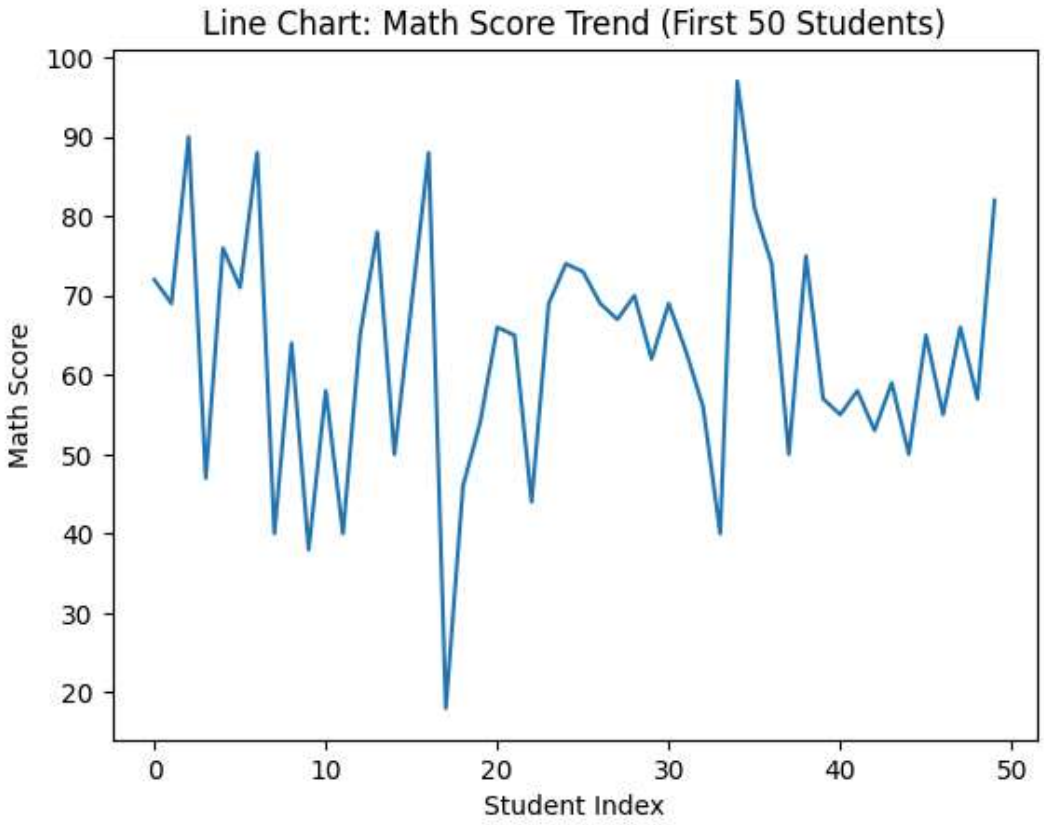
```
plt.xlabel("Student Index")
plt.ylabel("Math Score")
plt.show()
```



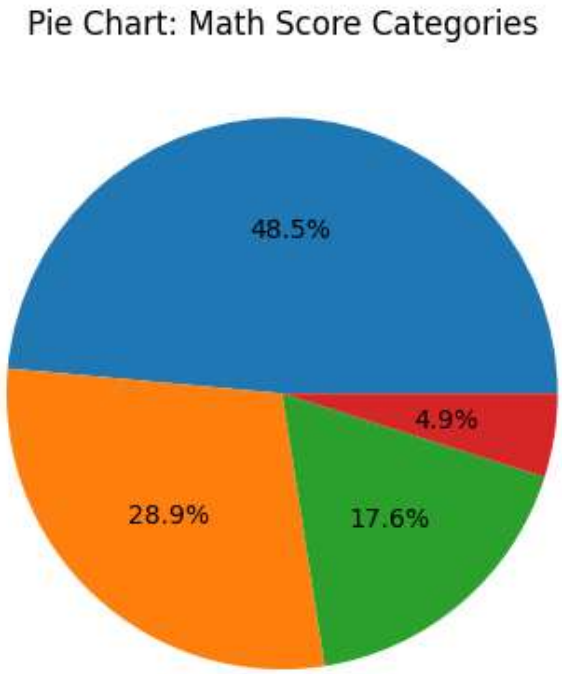

```
plt.figure()
plt.boxplot([df_csv['math_score'], df_csv['reading_score'], df_csv['writing_score']])
plt.title("Box Plot: Score Distributions")
plt.ylabel("Scores")
plt.show()
```




```
plt.figure()
plt.plot(df_csv['math_score'][:50])
plt.title("Line Chart: Math Score Trend (First 50 Students)")
plt.xlabel("Student Index")
plt.ylabel("Math Score")
plt.show()
```




```
categories = pd.cut(df_csv['math_score'], bins=[0,40,60,80,100],
                    labels=['Low','Average','Good','Excellent'])
pie_data = categories.value_counts()

plt.figure()
plt.pie(pie_data, autopct='%1.1f%%')
plt.title("Pie Chart: Math Score Categories")
plt.show()
```

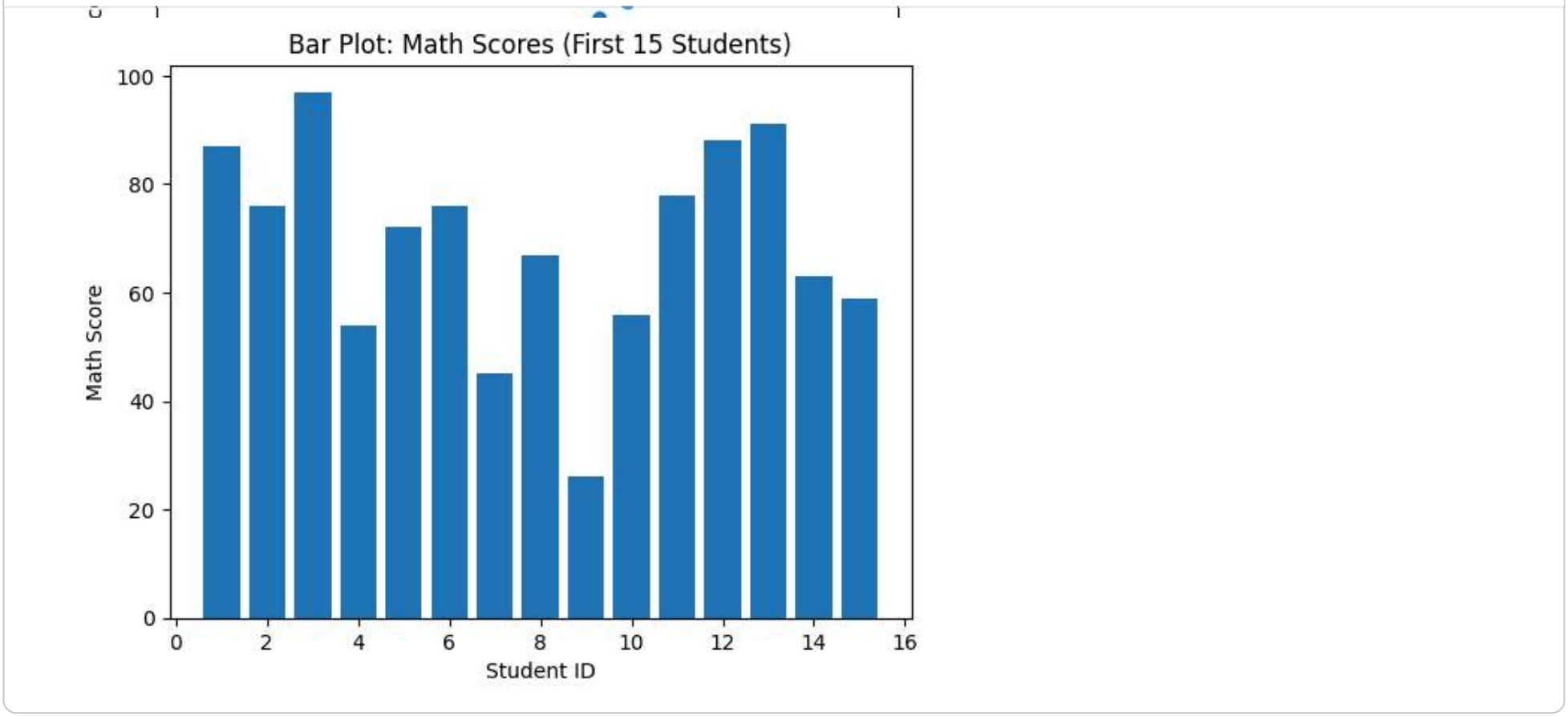



```
import pandas as pd
import matplotlib.pyplot as plt

student1 = pd.DataFrame({
    'Student_ID': list(range(1,31)),
    'Math_Score': [87,76,97,54,72,76,45,67,26,56, 78,88,91,63,59,84,73,69,90,47, 52,61,79,85,66,58,71,93,40,64],
    'Science_Score': [82,70,95,50,68,74,48,65,30,60, 75,86,89,61,55,80,70,66,88,45, 50,59,76,83,62,54,69,91,42,63],
    'English_Score': [85,72,96,53,70,75,46,66,28,58, 77,87,90,62,57,82,72,68,89,46, 51,60,78,84,64,56,70,92,41,65]
})

student1.head()
```

| | Student_ID | Math_Score | Science_Score | English_Score |
|---|---|---|---|---|
| **0** | 1 | 87 | 82 | 85 |
| **1** | 2 | 76 | 70 | 72 |
| **2** | 3 | 97 | 95 | 96 |
| **3** | 4 | 54 | 50 | 53 |
| **4** | 5 | 72 | 68 | 70 |

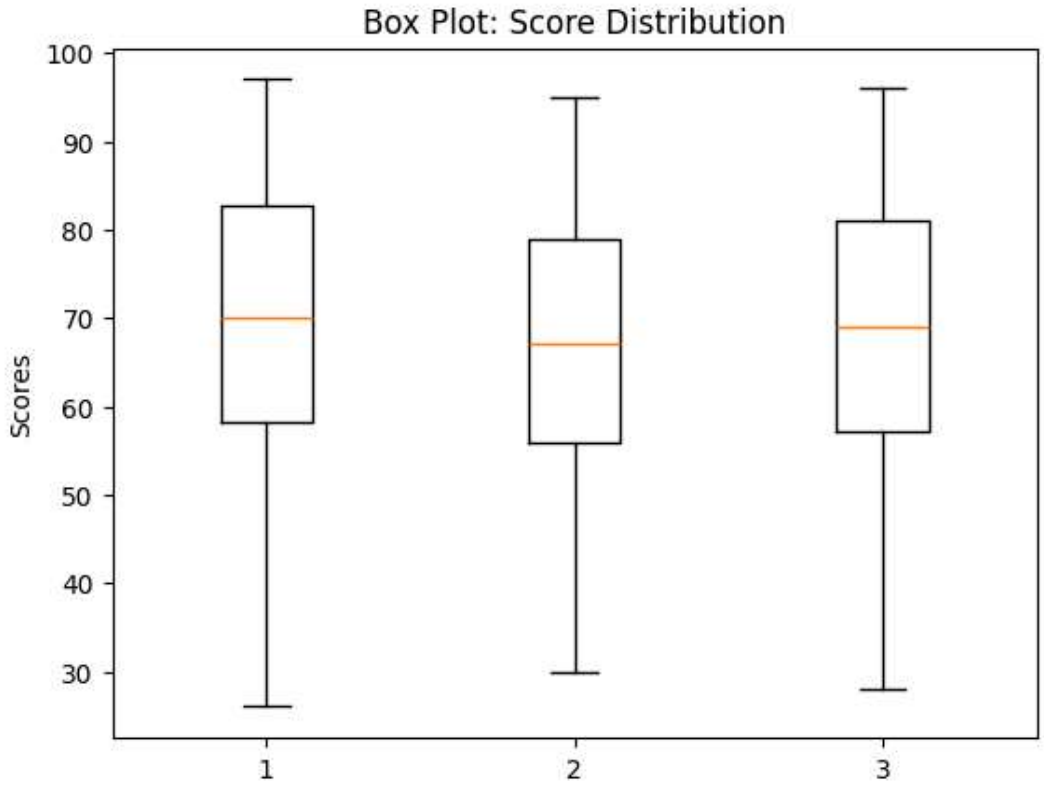
```
plt.figure()
plt.scatter(student1['Math_Score'], student1['Science_Score'])
plt.xlabel("Math Score")
plt.ylabel("Science Score")
plt.title("Scatter Plot: Math vs Science")
plt.show()
```

```
plt.figure()
plt.bar(student1['Student_ID'][:15], student1['Math_Score'][:15])
plt.xlabel("Student ID")
plt.ylabel("Math Score")
plt.title("Bar Plot: Math Scores (First 15 Students)")
plt.show()
```



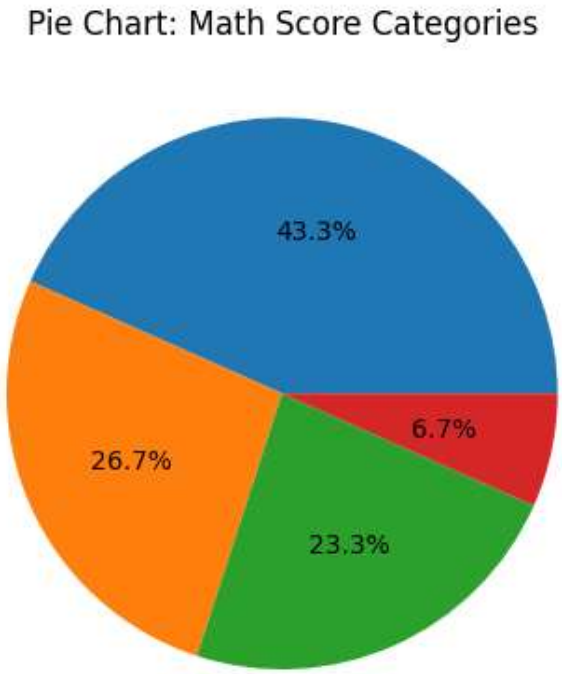

```
plt.figure()
plt.boxplot([
    student1['Math_Score'],
    student1['Science_Score'],
```

```
    student1['English_Score']
])
plt.title("Box Plot: Score Distribution")
plt.ylabel("Scores")
plt.show()
```

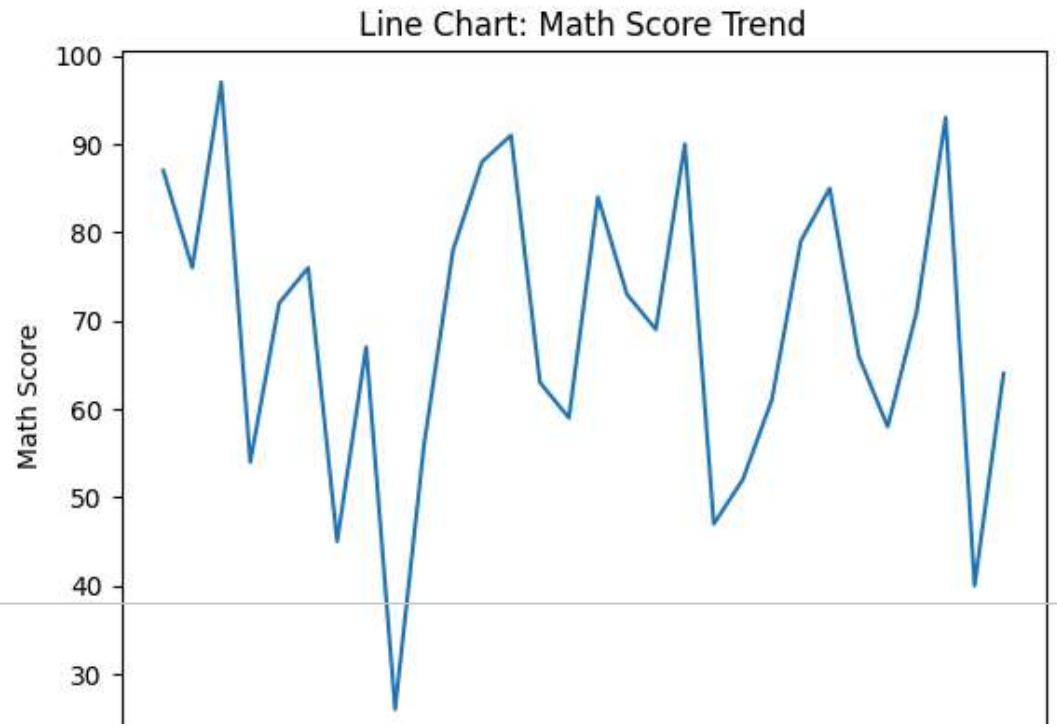



```
categories = pd.cut(student1['Math_Score'],
                    bins=[0,40,60,80,100],
                    labels=['Low','Average','Good','Excellent'])

pie_data = categories.value_counts()

plt.figure()
plt.pie(pie_data, autopct='%1.1f%%')
plt.title("Pie Chart: Math Score Categories")
```

```
plt.show()
```




```
plt.figure()
plt.plot(student1['Student_ID'], student1['Math_Score'])
plt.xlabel("Student ID")
plt.ylabel("Math Score")
plt.title("Line Chart: Math Score Trend")
plt.show()
```


Line Chart: Math Score Trend
Math Score
100
90
80
70
60
50
40
30